



Business Problem

An e-commerce company introduced a redesigned landing page with the objective of increasing user conversion rates. To evaluate whether the redesigned page performs better than the existing version, the company conducted an A/B testing experiment.

Users were randomly assigned into two groups:

- **Control Group** → shown the old landing page
- **Treatment Group** → shown the new landing page

The key business objective is to determine whether the new landing page generates a statistically significant improvement in conversion rate before making a full-scale deployment decision.

Main Objective

The primary objective of this analysis is to determine whether the new landing page leads to a higher conversion rate compared to the old landing page.

This project focuses on:

- Validating experiment data quality
- Comparing conversion performance between groups
- Performing statistical hypothesis testing
- Measuring confidence intervals
- Providing business recommendations based on statistical evidence

Import Libraries

```
In [ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")
plt.style.use("ggplot")
sns.set_palette("deep")
```

Import Dataset

```
In [ ]: df=pd.read_csv(r"ab_data.csv")
```

Understanding Dataset

```
In [ ]: df.head(5)
```

```
Out[ ]:
```

	user_id	timestamp	group	landing_page	converted
0	851104	2017-01-21 22:11:48.556739	control	old_page	0
1	804228	2017-01-12 08:01:45.159739	control	old_page	0
2	661590	2017-01-11 16:55:06.154213	treatment	new_page	0
3	853541	2017-01-08 18:28:03.143765	treatment	new_page	0
4	864975	2017-01-21 01:52:26.210827	control	old_page	1

```
In [ ]: df.tail(5)
```

```
Out[ ]:
```

	user_id	timestamp	group	landing_page	converted
294473	751197	2017-01-03 22:28:38.630509	control	old_page	0
294474	945152	2017-01-12 00:51:57.078372	control	old_page	0
294475	734608	2017-01-22 11:45:03.439544	control	old_page	0
294476	697314	2017-01-15 01:20:28.957438	control	old_page	0
294477	715931	2017-01-16 12:40:24.467417	treatment	new_page	0

```
In [ ]: df.shape
```

```
Out[ ]: (294478, 5)
```

```
In [ ]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 294478 entries, 0 to 294477
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   user_id          294478 non-null  int64
1   timestamp        294478 non-null  object
2   group            294478 non-null  object
3   landing_page     294478 non-null  object
4   converted        294478 non-null  int64
dtypes: int64(2), object(3)
memory usage: 11.2+ MB

```

```
In [ ]: df.describe()
```

```

Out[ ]:

```

	user_id	converted
count	294478.000000	294478.000000
mean	787974.124733	0.119659
std	91210.823776	0.324563
min	630000.000000	0.000000
25%	709032.250000	0.000000
50%	787933.500000	0.000000
75%	866911.750000	0.000000
max	945999.000000	1.000000

```
In [ ]: df.columns
```

```

Out[ ]: Index(['user_id', 'timestamp', 'group', 'landing_page', 'converted'], dtype='object')

```

Data Cleaning

Before performing statistical analysis, the dataset must be validated to ensure experiment integrity.

The following checks are performed:

- Missing value detection
- Duplicate record identification
- Validation of experiment group assignments
- Consistency between group and landing page mappings

This step is critical because incorrect experiment assignments can bias the A/B

testing results.

```
In [ ]: df.isnull().sum()
```

```
Out[ ]:
```

	0
user_id	0
timestamp	0
group	0
landing_page	0
converted	0

dtype: int64

```
In [ ]: df.duplicated().sum()
```

```
Out[ ]: np.int64(0)
```

Experiment Consistency Validation

In a valid A/B experiment:

- Users in the control group should only see the old page
- Users in the treatment group should only see the new page

Any mismatched assignments indicate experiment contamination and must be removed before analysis.

```
In [ ]: # Check whether users were assigned correctly to landing pages  
df.groupby(['group', 'landing_page']).size()
```

```
Out[ ]:
```

	0	
group	landing_page	
control	new_page	1928
	old_page	145274
treatment	new_page	145311
	old_page	1965

dtype: int64

```
In [ ]: # Filter dataset to retain only valid experiment assignments
```

```
df = df[
(df['group']=='control') & (df['landing_page']=='old_page') |
(df['group']=='treatment') & (df['landing_page']=='new_page')
]
```

```
In [ ]: # Verify that mismatched records were removed successfully
df.groupby(['group', 'landing_page']).size()
```

```
Out[ ]:
0
```

group	landing_page	
control	new_page	1928
	old_page	145274
treatment	new_page	145311
	old_page	1965

dtype: int64

Feature Engineering

Additional time-based features were extracted from the timestamp column to analyze behavioral trends across:

- Hour of the day
- Day of the week
- Daily traffic patterns

These features help identify whether user activity and conversion behavior vary over time.

```
In [ ]: # Convert timestamp column into datetime format
df['timestamp'] = pd.to_datetime(df['timestamp'])

# Extract day, hour, and weekday from timestamp
df['day'] = df['timestamp'].dt.day
df['hour'] = df['timestamp'].dt.hour
df['weekday'] = df['timestamp'].dt.day_name()
```

```
In [ ]: # Display updated dataset columns after feature engineering
df.columns
```

```
Out[ ]: Index(['user_id', 'timestamp', 'group', 'landing_page', 'converted', 'day',
              'hour', 'weekday'],
              dtype='object')
```

Exploratory Data Analysis (EDA)

```
In [ ]: # Count users in each experiment group
df['group'].value_counts()
```

```
Out[ ]:
```

	count
group	
treatment	147276
control	147202

dtype: int64

```
In [ ]: # Calculate percentage distribution across experiment groups
(df['group'].value_counts(normalize=True) * 100).round(2)
```

```
Out[ ]:
```

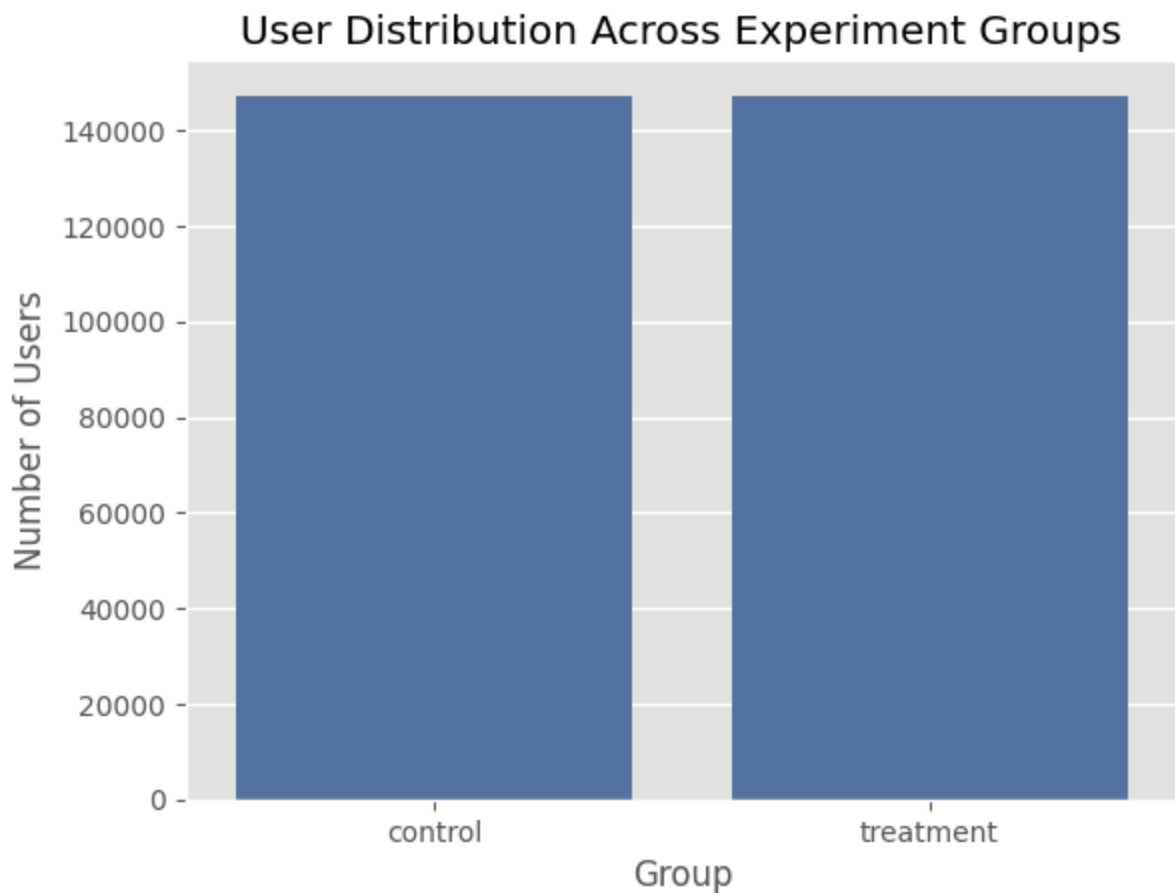
	proportion
group	
treatment	50.01
control	49.99

dtype: float64

```
In [ ]: sns.countplot(x='group', data=df)

plt.title('User Distribution Across Experiment Groups')
plt.xlabel('Group')
plt.ylabel('Number of Users')

plt.show()
```



Detailed Insights

- The visualization shows that users were distributed almost equally between the control and treatment groups.
- Balanced allocation indicates proper randomization during the experiment setup.

```
In [ ]: (df.groupby('group')['converted'].mean() * 100).round(2)
```

```
Out[ ]:          converted
```

group	
control	12.04
treatment	11.89

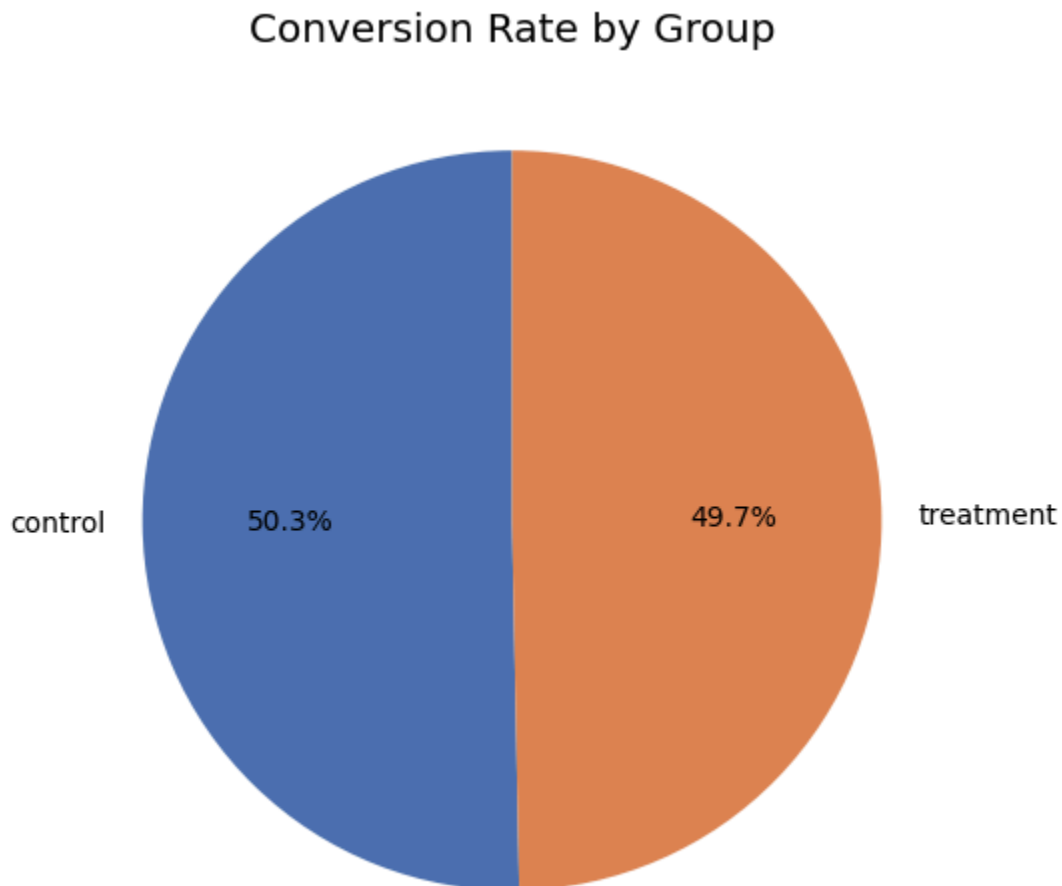
dtype: float64

```
In [46]: conversion_rate = df.groupby('group')['converted'].mean().reset_index()  
  
plt.figure(figsize=(6,6))
```

```
plt.pie(
    conversion_rate['converted'],
    labels=conversion_rate['group'],
    autopct='%1.1f%%',
    startangle=90
)

plt.title('Conversion Rate by Group')

plt.show()
```



Detailed Insights

- The conversion rates of the treatment and control groups appear very close to each other.
- There is no visually strong uplift in conversion for the new landing page.

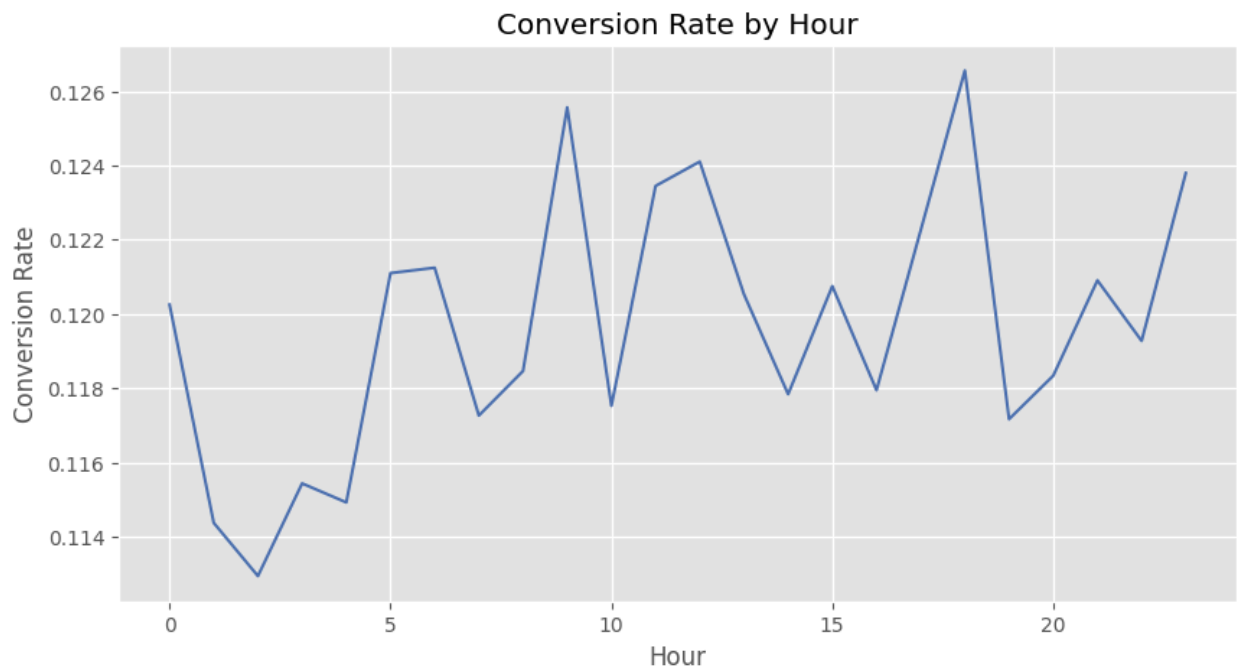
```
In [ ]: hourly_conversion = df.groupby('hour')['converted'].mean()

hourly_conversion.plot(kind='line', figsize=(10,5))
```



```
plt.title('Conversion Rate by Hour')
plt.ylabel('Conversion Rate')
plt.xlabel('Hour')

plt.show()
```



Detailed Insights

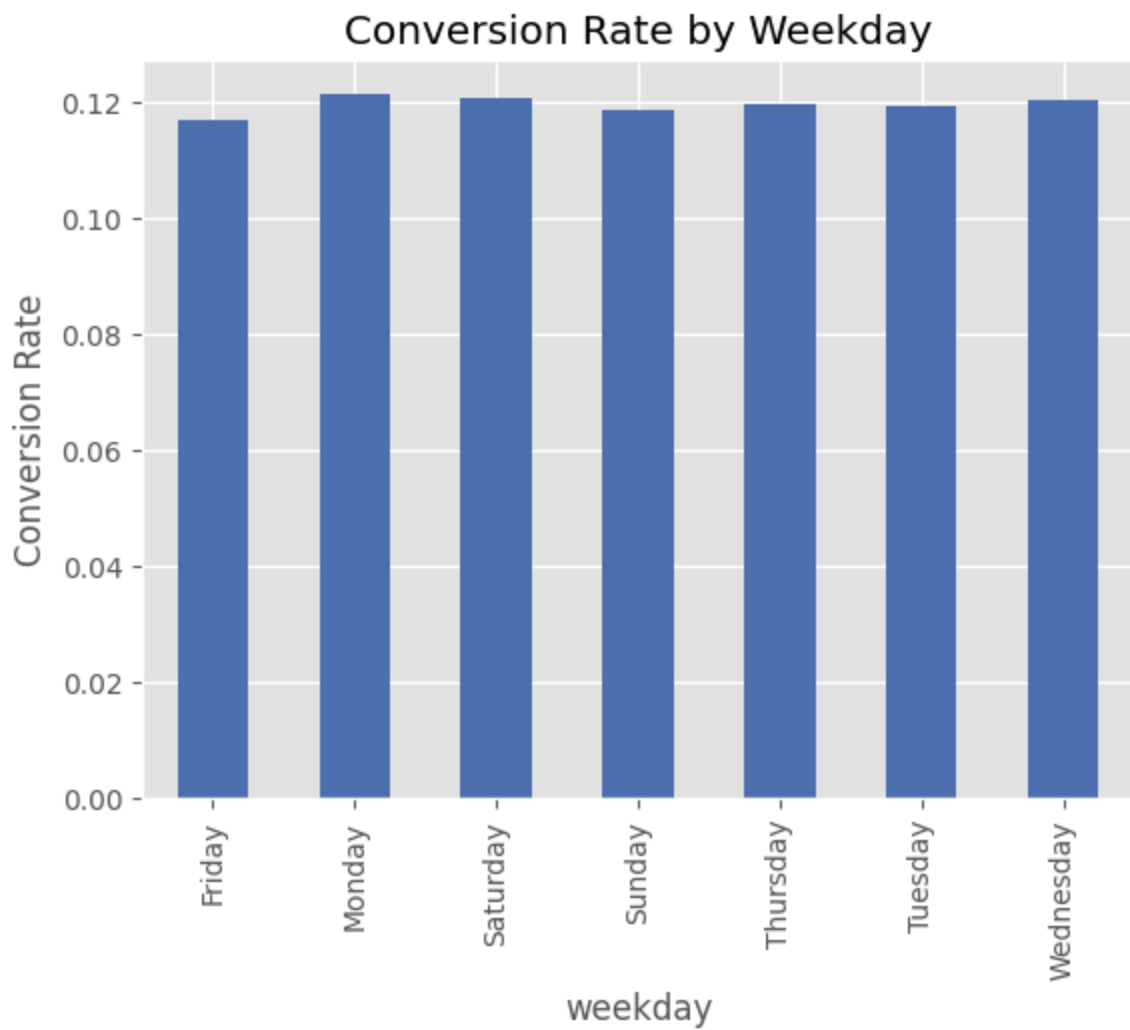
- Conversion rates fluctuate across different hours of the day.
- Certain time windows show relatively higher engagement and conversion activity.

```
In [42]: weekday_conversion = df.groupby('weekday')['converted'].mean()

weekday_conversion.plot(kind='bar')

plt.title('Conversion Rate by Weekday')
plt.ylabel('Conversion Rate')

plt.show()
```



Detailed Insights

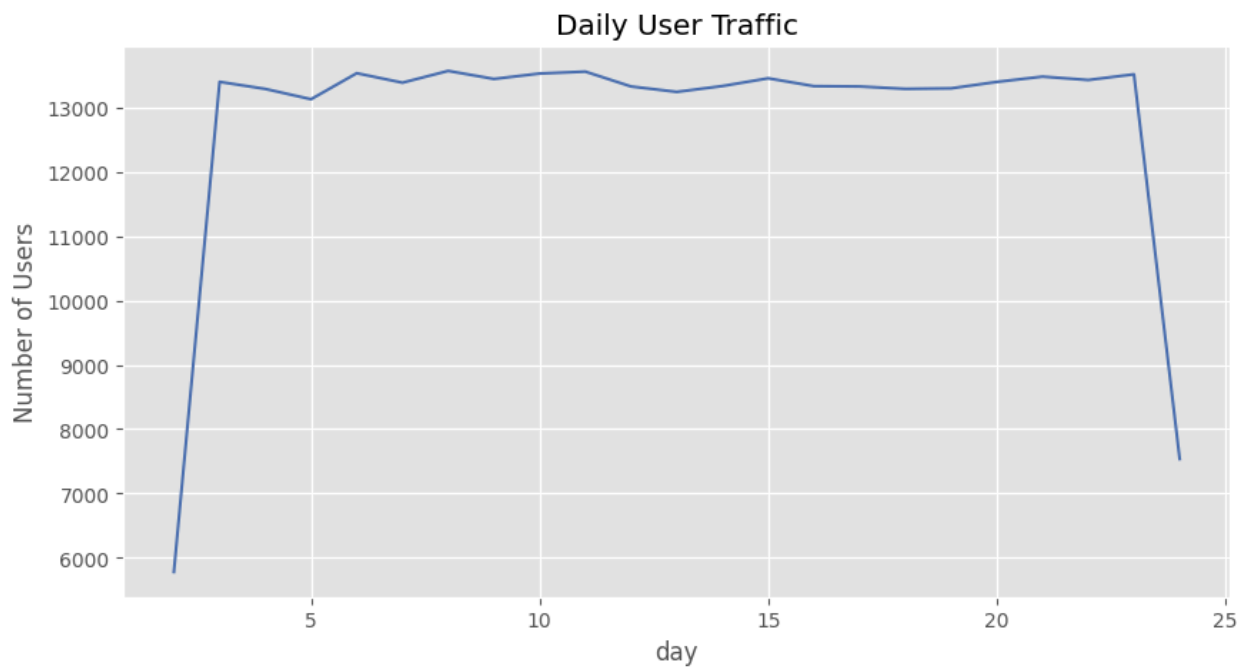
- Conversion performance varies slightly across weekdays.
- Some days consistently generate stronger conversion activity than others.

```
In [45]: daily_traffic = df.groupby('day').size()

daily_traffic.plot(kind='line', figsize=(10,5))

plt.title('Daily User Traffic')
plt.ylabel('Number of Users')

plt.show()
```



Detailed Insights

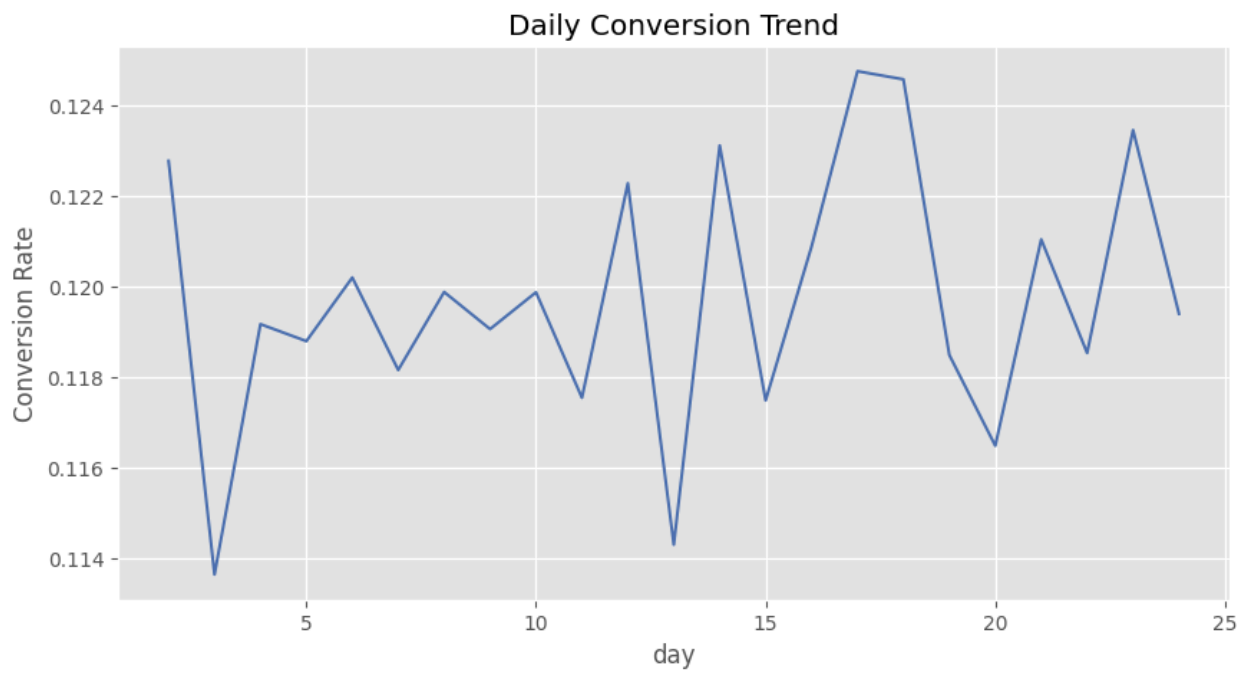
- Daily user traffic remains relatively stable throughout the experiment period.
- No major spikes or sudden drops are observed.

```
In [44]: daily_conversion = df.groupby('day')['converted'].mean()

daily_conversion.plot(kind='line', figsize=(10,5))

plt.title('Daily Conversion Trend')
plt.ylabel('Conversion Rate')

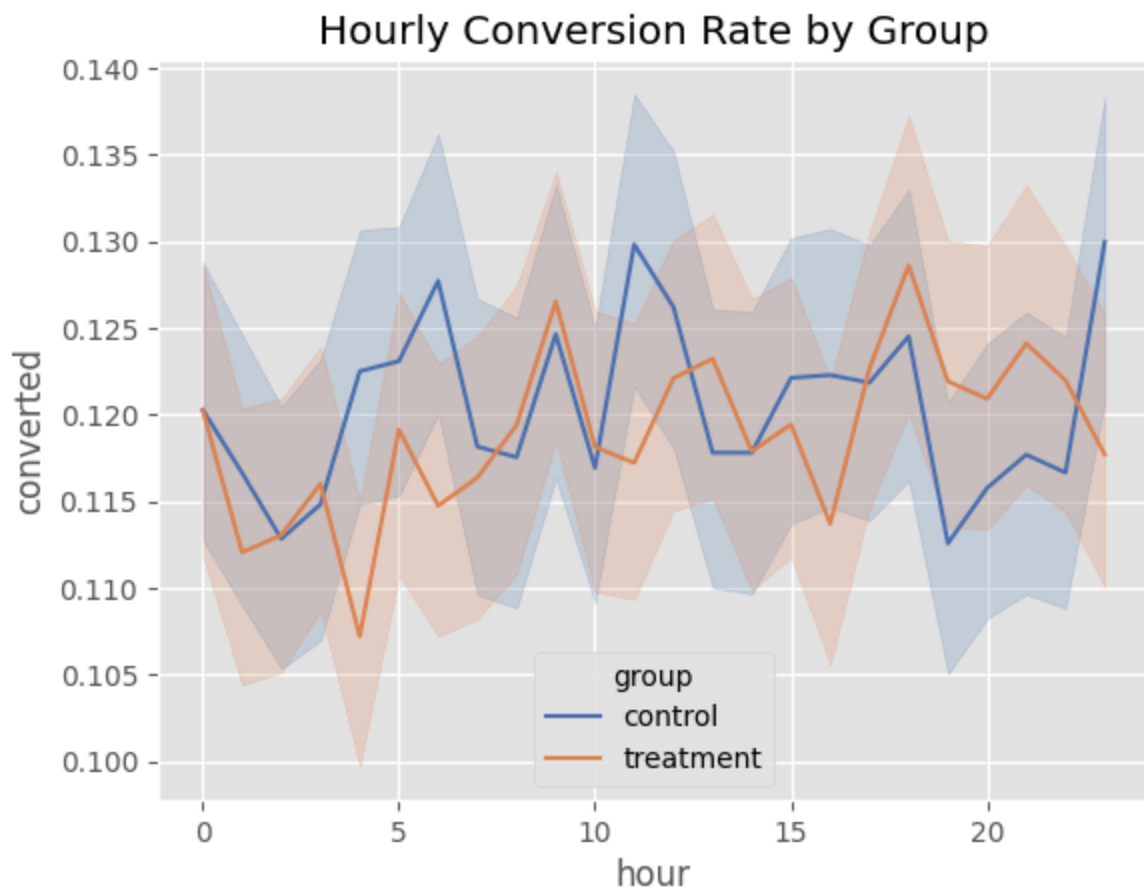
plt.show()
```



Detailed Insights

- Daily conversion rates fluctuate moderately across the experiment timeline.
- No strong upward or downward trend is observed.

```
In [43]: sns.lineplot(data=df, x='hour', y='converted', hue='group')  
plt.title('Hourly Conversion Rate by Group')  
plt.show()
```



Detailed Insights

- Both control and treatment groups follow very similar hourly conversion patterns.
- No consistent time period shows strong dominance by the treatment group.

Hypothesis Testing

The objective of the hypothesis test is to determine whether the new landing page produces a higher conversion rate than the existing landing page.

Null Hypothesis (H_0)

The new landing page does not improve the conversion rate.

Alternative Hypothesis (H_1)

The new landing page improves the conversion rate.

A one-tailed two-proportion Z-test is used because the objective is specifically to test whether the new page performs better than the old page.

Statistical Assumptions

The two-proportion Z-test assumes:

- Independent observations
- Random assignment of users
- Sufficiently large sample sizes
- Binary conversion outcomes

Given the large experiment size, the assumptions required for the Z-test are reasonably satisfied.

```
In [ ]: # Extracts required metrics for the two-proportion Z-test.
control_group = df[df['group'] == 'control']
treatment_group = df[df['group'] == 'treatment']

control_converted = control_group['converted'].sum()
treatment_converted = treatment_group['converted'].sum()

control_size = control_group.shape[0]
treatment_size = treatment_group.shape[0]
```

```
In [ ]: # Tests whether the new landing page performs significantly better than the old
from statsmodels.stats.proportion import proportions_ztest

conversions = [treatment_converted, control_converted]
sample_sizes = [treatment_size, control_size]

z_stat, p_value = proportions_ztest(conversions, sample_sizes, alternative='less')

print("Z-statistic:", z_stat)
print("P-value:", p_value)
```

```
Z-statistic: -1.2369217547321678
P-value: 0.8919419336512124
```

Interpretation

- A low p-value (< 0.05) would indicate statistical evidence that the new landing page performs better.
- A high p-value suggests insufficient evidence to conclude that the new page improves conversion rates.

The observed p-value indicates that the conversion difference is not statistically significant.

Confidence Interval

```
In [ ]: # Calculate confidence interval for difference in conversion rates
from statsmodels.stats.proportion import confint_proportions_2indep

low_ci, high_ci = confint_proportions_2indep(
    count1=treatment_converted,
    nobs1=treatment_size,
    count2=control_converted,
    nobs2=control_size,
    method='wald'
)

print("Confidence Interval:")
print(low_ci, high_ci)
```

Confidence Interval:
-0.003824096247825661 0.0008648966596705573

Confidence Interval Interpretation

The confidence interval estimates the likely range of the true difference in conversion rates between the treatment and control groups. Since the interval includes zero, the observed difference may be due to random variation rather than a genuine improvement from the new landing page.

Final Conclusion

The A/B testing analysis indicates that the redesigned landing page did not produce a statistically significant improvement in conversion rate compared to the existing landing page.

Key findings from the experiment include:

- Conversion rates between both groups were very similar
- The p-value from the two-proportion Z-test exceeded the significance threshold of 0.05
- The confidence interval included zero, suggesting that the observed difference may be attributable to random chance

Based on the statistical evidence, there is insufficient support to conclude that the new landing page outperforms the current version.

Business Recommendation

Based on the experimental results, deploying the new landing page is not currently recommended because the observed improvement in conversion rate is not statistically significant.

Recommended next steps:

- Retain the existing landing page
- Conduct additional user behavior analysis
- Investigate alternative landing page designs
- Perform segmented analysis across device type, traffic source, or user demographics
- Run future experiments with stronger design variations

The current redesign may not provide enough user experience improvement to meaningfully influence conversion behavior.

Limitations

This analysis has several limitations:

- Only overall conversion rate was analyzed
- User-level behavioral factors were not considered
- External influences such as traffic quality and marketing campaigns were not evaluated
- Long-term customer retention impact was not measured

Future experiments could incorporate deeper segmentation and behavioral analytics.